



MARKING SCHEME MATHEMATICS FORM 2

1. Use tables to evaluate.

(3mks)

$$\sqrt{\left(\frac{3.45 \times 16.7}{31.5} \right)}$$

NUMBER	STD FORM	LOG
3.45	3.45×10^0	0.5378
16.7	1.67×10^1	1.22227 ⁻¹
		1.7605
31.5	3.15×10^1	1.4983-
		0.2622 x1/2
1.3524	1.35×10^0	0.1311
=1.3526		

2. Solve for x in each of the following equations.

(3mks)

(a) $3^{(2x-5)} = 3^3$

$$2x-5=3$$

$$\underline{2x=8}$$

$$\underline{2 \quad 2}$$

$$X=4$$

(b) $3^{4x} \div 3^{-7} = 3^{15}$

(3mks)

$$3^{4x-7} = 3^{15}$$

$$3^{4x+7} = 3^{15}$$

$$4x+7=15$$

$$4x=15-7$$

$$\underline{4x=8}$$

$$\underline{4 \quad 4}$$

$$X=2$$

3. Use reciprocals tables to evaluate

(3mks)

$$\frac{7}{0.0125} + \frac{1}{12.5}$$

$$\left(\frac{1}{0.0125} \right) + 1/12.5$$



$$560 \times 7 + 0.08$$

$$3920 + 0.08$$

$$3920.08$$

4. A metallic cuboid measuring 16cm by 8cm by 4 cm was melted . The material was then used to make a cube. What was the length of the cube? (3mks)

$$16 \times 8 \times 4 = 512 \text{ cm}^3$$

$$\sqrt[3]{512} = 8 \text{ cm}$$

5. Simplify

$$\sqrt[3]{\frac{27x^3y^9}{x^6y^3}}$$

(3mks)

$$\frac{3xy^3}{x^2y}$$

$$\frac{3y^2}{x}$$

$$\frac{3y^2}{x}$$

$$\text{Or } 3y^2x^{-1}$$

6. Find the equation of the line through the points A (2, 5) and B(3, 11)

(3mks)

$$M = \frac{11-5}{3-2} = \frac{6}{1}$$

$$= 6$$

$$\frac{Y-5}{X-2} = 6$$

$$A(2, 5) \quad C(x, 7)$$

$$\frac{y-5}{x-2} = 6$$

$$x-2$$

$$y-5 = 6(x-2)$$

$$y-5 = 6x-12$$

$$y = 6x-7$$

$$y = 6x-7$$

7. Determine the equation of the line perpendicular to the line whose equation is $y = -5x + 3$ and passes through the point (3, 2). (3mks)

$$y = -5x + 3$$

$$m_1 = -5$$

$$m_1 \times m_2 = -1$$

$$-5 / -5m_2 = 1/5$$

$$m_2 = 1/5$$

$$\frac{y-2}{x-3} = \frac{1}{5}$$

$$x-3 = 5$$

$$5(y-2)=x-3$$

$$5y-10=x-3$$

$$5y=x-3+10$$

$$5y=x+7$$

$$Y=1/5x+7$$

8. A(-5, -2), B(-2, -5) and C(-12, -2) are vertices of a triangle. Find the image of the triangle when it is reflected in :

(a) The line $y=-x$

(4mks)

A i(4,2.5)

B' (10,1)

C '(4,6)

(b) The line $y= x$

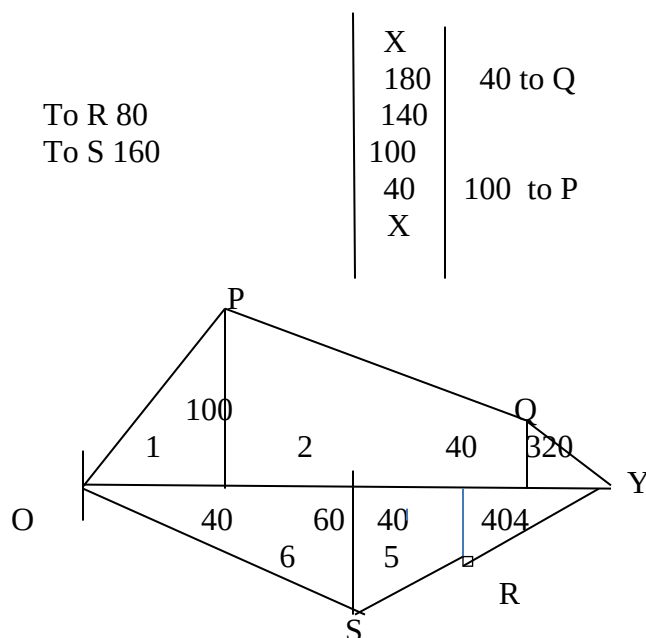
(4mks)

A "(-4, -2.5)

B "(-12, -2)

C "(-2, -5)

9. Find the area in hectares of a coffee filed whose measurements are entered in a filed book as shown below. Take $xy=200m$ as the baseline. (8mks)



$$\text{Area 1} = \frac{1}{2} \times 40 \times 100 = 2000m^2$$

$$\text{Area 2} = \frac{1}{2} (100+40)40 = 9000m^2$$

$$\text{Area 3} = \frac{1}{2} \times 20 \times 40 = 400m^2$$

$$12200m^2$$

$$\text{Area 4} = \frac{1}{2} \times 60 \times 80 = 2400m^2$$

$$\text{area} = \frac{1}{2} (160 + 80$$

$$0 \times 40 = 4800m^2$$

Area



10. Use the reciprocal tables and square root to evaluate.

(4mks)

$$\frac{0.1}{0.0351} + \sqrt{0.498}$$

0.0351

$$\left(\frac{1}{0.0351} \right) 0.1 + \sqrt{49.8 \times 10^{-2}}$$

$$(28.490)0.1 + \sqrt{49.8} \times \sqrt{10^{-2}}$$

$$2.849 + 7.057 \times 10^{-1}$$

$$2.849 + 0.7057$$

$$= 3.5547$$

11. Two men each working for 8 hours a day, can cultivate an acre of land in 4 days. How long would 6 men each working in 4 hours a day take to cultivate 4 acres? (3mks)

Men	Hours	Days	acres
2	8	4	1
6	4	?	4
2=4	2/6x44x8/4x4/1		
24/3=8 days			

12. The sum of interior angles of a regular polygon is 1080° . Find the size of each exterior angle. (3mks)

$$(2n-4)90$$

$$(2n-4)90=1080$$

$$180n-360=1080$$

$$180n=1080+360$$

$$\underline{180n=1440}$$

$$180 \quad 180$$

$$N=8$$

$$1080/8=135$$

$$180-135=45^\circ$$

Or

$$360/8=45^\circ$$